Research Positioning and Scientific Contribution

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June 9, 2026

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Research Context

Climate systems are characterized by strong spatial and temporal variability arising from complex interactions between atmospheric, hydrological, and environmental processes. Classical deterministic partial differential equations are often insufficient to represent uncertainties associated with unresolved physical processes, measurement errors, and random environmental perturbations.

Stochastic partial differential equations provide a natural mathematical framework for incorporating such uncertainties into climate and environmental models.

Problem Considered

We consider a stochastic advection–diffusion equation of the form

$$du = (-\mathbf{b} \cdot \nabla u + \nabla \cdot (\kappa \nabla u) + f)\, dt + \sigma \phi(x), dW_t,$$

where

- $u(x,t)$ represents a climatic variable (temperature, humidity, pollutant concentration, etc.);
- $\mathbf{b}(x,t)$ denotes a transport velocity field;
- κ is the diffusion coefficient;
- f is a source term;
- $\phi(x)$ describes the spatial structure of the uncertainty;
- W_t is a Wiener process.

Scientific Motivation

Many environmental and climate models involve advection and diffusion mechanisms subject to uncertain forcing. Examples include:

- atmospheric temperature transport;
- humidity dynamics;
- contaminant transport;
- dispersion of greenhouse gases;
- uncertainty propagation in climate simulations.

Although stochastic diffusion equations have been extensively studied, stochastic advection–diffusion equations remain comparatively less explored, especially from the viewpoint of finite element approximation and convergence analysis.

Research Gap

Existing literature mainly focuses on:

- stochastic diffusion equations;
- reaction–diffusion SPDEs;
- abstract stochastic evolution equations.

Fewer studies address the simultaneous treatment of

- advection;
- diffusion;
- stochastic forcing;
- finite element approximation;
- climate-oriented applications.

Main Contributions

The proposed work aims to provide:

1. a variational formulation of a stochastic advection–diffusion equation;
2. stability and energy estimates;
3. existence and uniqueness results;
4. a finite element approximation based on continuous piecewise linear elements;
5. a convergence analysis of the fully discrete scheme;
6. numerical experiments illustrating uncertainty propagation in transport phenomena;
7. a framework suitable for future climate-related applications.

Long-Term Perspective

The present work constitutes a first step towards the development of stochastic finite element models for climate and environmental systems.

Future extensions include:

- multiplicative noise;
- random diffusion coefficients;
- stochastic fluid dynamics;
- data assimilation;
- climate reanalysis data integration (ERA5);
- uncertainty quantification for environmental prediction.